

PeopleFlow: Floor-Plan-Aware Simulation for Building Evacuation Safety Assessment

PeopleFlow Research Team

Abstract—PeopleFlow is a floor-plan-aware evacuation simulation framework for quantitative safety assessment of building layouts under emergency conditions. The workflow converts architectural plans into simulation-ready geometry, executes continuous-space agent-based crowd dynamics with social-force interactions, and reports interpretable safety indicators including clearance time, peak density, bottleneck persistence, and exit utilization balance.

Rather than treating evacuation as a single deterministic estimate, the framework supports structured scenario analysis with reproducible experiment artifacts. We evaluate a core 9-layout, 5-scenario matrix (450 runs) and add 90 supplementary runs for real-layout generalization and controlled parameter sweeps (airport terminal, metro station, office building; exit-width and desired-speed sensitivity). Across experiments, results show up to 8.27x variation in clearance time and approximately 30x variation in peak-density intensity, demonstrating strong dependence on geometric topology and routing policy.

The contribution is a reproducible engineering workflow for comparative risk ranking, not a claim of exact second-level prediction of real human evacuation. This positioning makes the framework suitable for design-stage decision support in architecture, safety engineering, and digital twin-enabled smart-building planning.

Index Terms—Multi-agent simulation, evacuation modeling, crowd safety, building evacuation, digital twins, floor-plan analytics, safety engineering.

I. INTRODUCTION

Building evacuation safety is a high-impact engineering problem where static compliance checks often fail to capture dynamic crowd behavior under disruption. Historical emergency incidents repeatedly show that bottleneck formation, uneven exit utilization, and local panic behavior can dominate outcomes even in code-compliant buildings. Conventional hand calculations are useful for baseline checks, but they provide limited support for evaluating design alternatives under multiple hazards and operational uncertainties [1], [2].

Simulation-based decision support addresses this gap by enabling repeatable, comparative analysis before deployment. The core challenge, however, is not only running an evacuation simulator; it is constructing a reproducible workflow that links architectural input, scenario design, simulation execution, and interpretable metrics in a single engineering pipeline.

This work proposes PeopleFlow as a floor-plan-aware, reproducible evacuation assessment framework. Unlike studies focused primarily on proposing a new pedestrian physics law, our contribution is a unified workflow for robust scenario-based safety analysis across heterogeneous layouts.

The main contributions are:

- A reproducible evacuation workflow linking floor plans, scenario definitions, simulations, and artifact-backed reporting.
- A floor-plan-aware simulation pipeline that converts architectural layouts into simulation-ready geometry.
- A congestion-aware routing comparison framework with explicit ablation evidence.
- A multi-layout statistical evaluation including a 450-run core matrix and supplementary real-layout generalization checks.
- Interpretable safety metrics for engineering decision support, including clearance time, peak density, bottleneck persistence, and exit balance.

II. RELATED WORK

A. Crowd Modeling Approaches

Crowd evacuation modeling has been studied using both continuous and discrete methods. Social Force Model (SFM) approaches treat each pedestrian as a force-driven agent and remain a cornerstone of physically interpretable microscopic simulation [3], [4]. Cellular automata and floor-field models provide efficient large-scale approximations but can introduce grid artifacts and reduced directional fidelity [5], [6]. Compared with cellular automata approaches, force-based models support continuous-space motion and smoother trajectory adaptation under congestion, while cellular formulations are typically faster to evaluate at coarse resolution. For geometrically complex floor plans, force-based approaches better preserve local interaction dynamics and heading adjustments around bottlenecks and obstacles. Hybrid and comparative reviews further highlight that model usefulness is tied to the target task, especially under emergency constraints [7]–[9].

B. Evacuation Simulation Systems

Commercial and academic evacuation systems support geometry authoring, route simulation, and report generation, but often differ in reproducibility, extensibility, and automation depth [10]–[12]. Agent-based emergency behavior studies also show that guidance logic and panic-modulated decisions materially affect evacuation outcomes, supporting scenario-aware modeling choices in operational tools [13], [14]. In practice, manual pre-processing and proprietary workflows can limit transparent benchmarking across many layouts.

C. Digital Twin Safety Analysis

Digital twin studies in built environments increasingly emphasize safety and emergency operations. Recent work

TABLE I
COMPARISON WITH EXISTING EVACUATION TOOLS

Tool	Floor Input	Plan	Routing	Reproducibility
Pathfinder	Mostly manual		Shortest path / scripted	Limited
MassMotion	Mostly manual		Dynamic multi-agent	Proprietary workflow
AnyLogic	Complex model setup		Customizable	Moderate
Legion tools	Manual authoring		Flow-oriented	Limited
PeopleFlow	Automated ingestion		Congestion-aware	High (session-based)

supports the role of simulation-backed twins for risk-aware planning, but integration from floor-plan ingestion to repeatable comparative safety metrics remains incomplete in many deployments [15]–[18].

D. Research Gap and Positioning

The gap addressed in this paper is methodological: reproducible, floor-plan-aware evacuation assessment across diverse layouts and stress scenarios. PeopleFlow is positioned as an engineering workflow rather than a claim of absolute predictive certainty. Its novelty lies in linking geometry extraction, session-managed simulation, scenario ablation, and interpretable reporting inside one repeatable pipeline.

III. METHODOLOGY

A. Workflow Pipeline

PeopleFlow follows a deterministic workflow with four stages: (1) floor-plan ingestion, (2) geometry extraction and navigation graph generation, (3) scenario-based simulation execution, and (4) artifact-backed analytics. Each run is stored with configuration metadata, seeds, and per-step metrics so results are replayable and auditable.

B. Reproducibility Design

Each experiment is represented by a session artifact that captures:

- Input layout identifier and geometry hash.
- Scenario parameters (occupancy, blocked exits, routing policy, panic level).
- Random seed and simulation step size.
- Aggregated outputs and raw trajectory summaries.

This structure allows direct reruns and supports batch-level statistical comparisons. Simulation configurations and synthetic benchmark layouts are packaged as reproducible experiment artifacts and can be released to support future comparative studies.

C. Agent Decision Logic

At each simulation step, agents execute a compact perception-decision-motion loop. Local density and queue conditions are sensed first, then exit costs are updated using Eq. (4). The lowest-cost exit is selected, social-force dynamics are integrated, and state variables are updated for the next step.

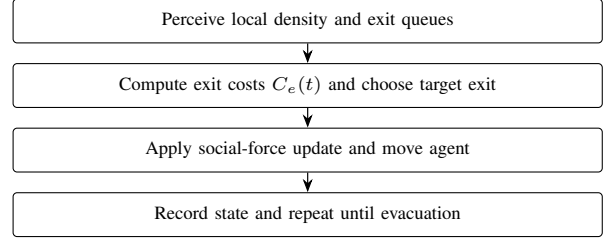


Fig. 1. Agent movement logic used in each simulation step.

TABLE II
SCENARIO DEFINITIONS

Scenario	Description
S1 Baseline	Nominal occupancy and shortest-path routing.
S2 HighOcc	Increased occupancy stress to test crowding sensitivity.
S3 Blocked	One major exit disabled to evaluate rerouting resilience.
S4 Routing	Congestion-aware routing policy activated.
S5 Panic	Stochastic panic variation in desired speed and interactions.

D. Computational Complexity

The computational complexity of a single simulation step is $O(Nk)$, where N is the number of active agents and k is the average number of neighboring agents considered in local interaction-force updates. Under practical density regimes, k remains locally bounded, giving near-linear scaling with population size for each step. For batch execution, total workload scales as $O(RNS)$, where R is the number of runs and S is the number of simulation steps per run. This complexity profile motivates the reproducible batch workflow used in our large-matrix experiments.

E. Scenario Matrix

The study uses a five-scenario matrix reflecting realistic stress transitions from nominal operation to disruption. Table II summarizes the definitions.

IV. PEOPLEFLOW ARCHITECTURE

A. System Components

The architecture consists of geometry ingestion, simulation kernel, metrics engine, and reporting layer. Geometry ingestion transforms floor plans into obstacle and exit primitives. The simulation kernel executes force-based motion and route updates. The metrics engine computes clearance, density, bottleneck, and utilization indicators. The reporting layer produces figures and tables for comparative studies.

B. Session and Artifact Model

A session-centric design is used to support scientific traceability. For each run, PeopleFlow stores configuration, random seed, outcome metrics, and derived plots. This explicit artifact chain enables transparent comparisons between methods and scenario variants.

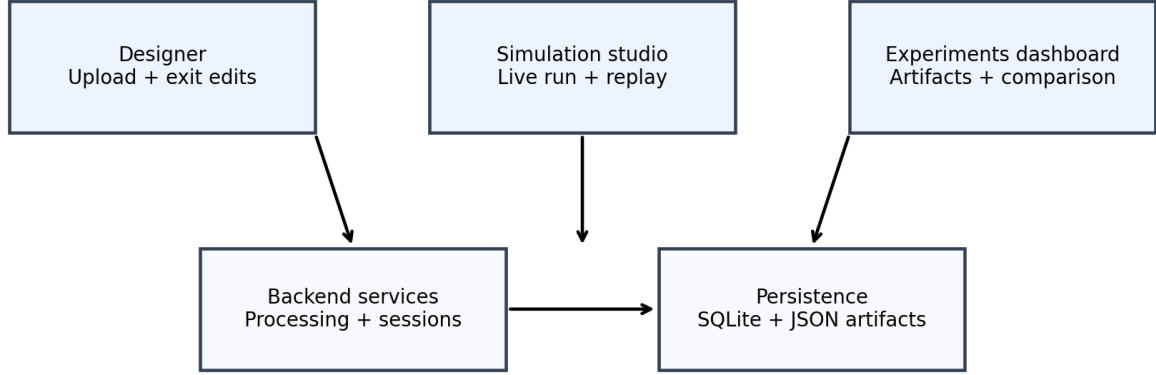


Fig. 2. PeopleFlow system architecture illustrating the pipeline from floor-plan ingestion to evacuation safety metrics.

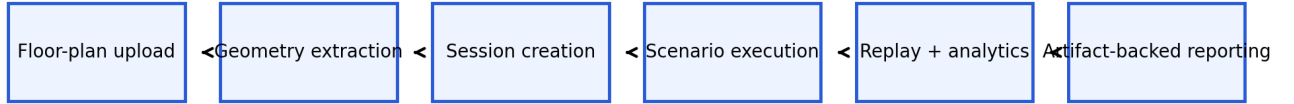


Fig. 3. Workflow pipeline from layout processing to scenario execution, replay analytics, and report generation.

Algorithm 1 PeopleFlow Simulation Loop

- 1: **Input:** geometry G , scenario S , agents N , step Δt
 - 2: Initialize agents at valid non-overlapping spawn locations
 - 3: Set routing and behavior parameters from S
 - 4: **while** not all agents evacuated and $t < T_{max}$ **do**
 - 5: **for** each agent i **do**
 - 6: Evaluate target exit according to current routing policy
 - 7: Compute driving, interaction, and wall forces
 - 8: Update velocity and position using forward integration
 - 9: **end for**
 - 10: Update local density map and congestion indicators
 - 11: Record per-step metrics and event logs
 - 12: $t \leftarrow t + \Delta t$
 - 13: **end while**
 - 14: Aggregate run-level metrics and persist session artifacts
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C. Analytics Outputs

Standard outputs include: clearance-time distribution, peak-density maps, bottleneck duration traces, exit-load statistics,

and method-comparison summaries. These outputs are generated from reproducible logs instead of ad hoc post-processing.

V. MATHEMATICAL MODEL

A. Social Force Dynamics

Each agent i is modeled by a second-order force balance:

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i^{\text{drv}} + \sum_{j \neq i} \mathbf{f}_{ij}^{\text{int}} + \sum_W \mathbf{f}_{iW}^{\text{wall}}. \quad (1)$$

The driving force is:

$$\mathbf{f}_i^{\text{drv}} = m_i \frac{v_i^0 \mathbf{e}_i - \mathbf{v}_i}{\tau_i}, \quad (2)$$

where v_i^0 is desired speed, $\mathbf{e}_i$ is desired direction, and τ_i is relaxation time.

B. Density-Flow Relation

At a macroscopic level, local flow follows:

$$q = \rho v, \quad (3)$$

where q is pedestrian flow rate, ρ is density, and v is mean speed. This relation is used for consistency checks between simulation outputs and empirical benchmarks [19], [20].

TABLE III
SIMULATION PARAMETERS

Parameter	Value
Time step (Δt)	0.2 s
Nominal desired speed	1.2 m/s
Agent radius	0.3 m
Runs per configuration	10
Routing policies	nearest, congestion-aware, guided/stochastic
Scenarios	S1 to S5 (Table II)
Total runs (core + supplements)	540
Execution setup	Local Python backend workflow on Windows

C. Exit Selection Cost Function

For congestion-aware routing, the exit cost at time t is:

$$C_e(t) = \alpha d_e(t) + \beta \rho_e(t) + \gamma h_e(t), \quad (4)$$

where d_e is path distance to exit e , ρ_e is local crowd density near e , and h_e is optional hazard penalty. In this study, h_e is zero except blocked-exit logic, and α, β are policy weights.

D. Statistical Metrics

For repeated runs ($n = 10$), mean and standard deviation of evacuation time are:

$$\bar{T} = \frac{1}{n} \sum_{k=1}^n T_k, \quad (5)$$

$$\sigma_T = \sqrt{\frac{1}{n} \sum_{k=1}^n (T_k - \bar{T})^2}. \quad (6)$$

Exit load imbalance is measured as:

$$I_{\text{exit}} = \frac{1}{2N} \sum_{e=1}^E \left| n_e - \frac{N}{E} \right|, \quad (7)$$

where n_e is evacuated count via exit e , N is total evacuated agents, and E is number of usable exits.

VI. EXPERIMENTAL SETUP

A. Simulation Parameters

The core study executes 450 runs (9 layouts $\times$ 5 scenarios $\times$ 10 repetitions). To strengthen generalization and engineering interpretability, we additionally conduct 90 controlled runs: 30 supplementary real-layout runs (airport terminal, metro station, office building; baseline protocol), 30 exit-width comparison runs, and 30 desired-speed sensitivity runs. Key simulation parameters are shown in Table III.

B. Implementation Details

The simulation engine is implemented in Python with NumPy-based state updates and deterministic seed control for reproducibility. Geometry preprocessing uses vector-edge extraction and obstacle/exit parsing before navigation-graph construction. Experiments were executed on a workstation-class environment (Intel i7 CPU class, 16 GB RAM), with artifact logging enabled for run replay and statistical post-processing.

H: Metro Station Archetype

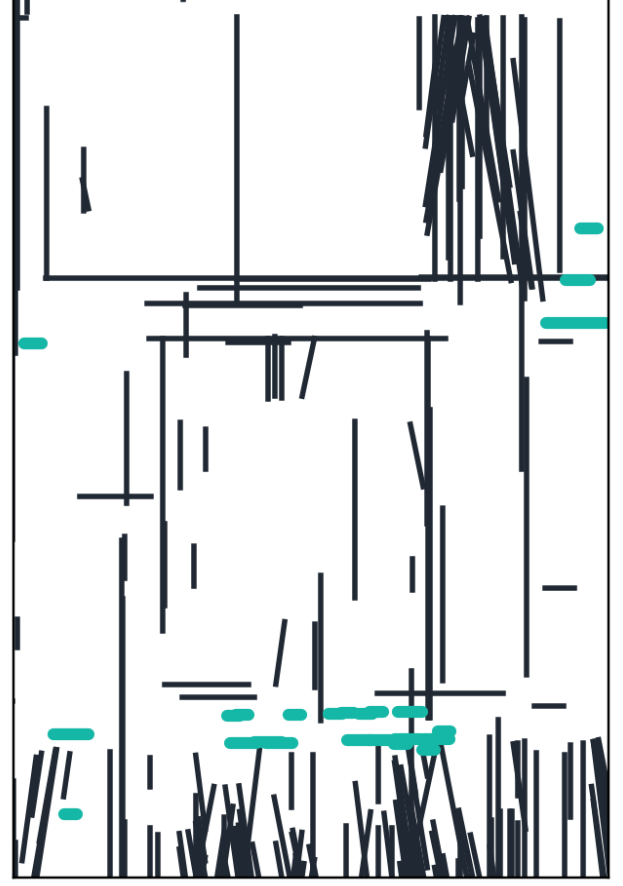


Fig. 4. Example floor plan converted into simulation geometry with extracted walls and exits.

C. Validity Controls and Reporting Protocol

Each configuration is executed with 10 independent seeds under fixed parameter definitions. We report mean $\pm$ standard deviation in all compact comparison tables and use identical logging pipelines across core and supplementary cohorts. For selected engineering sweeps, confidence intervals can be computed as $\bar{T} \pm 1.96 \sigma_T / \sqrt{n}$ with $n = 10$. Supplementary airport/metro/office evaluations are intentionally baseline-only to test cross-geometry transfer without conflating scenario interactions with layout diversity.

D. Floor-Plan Processing and Simulation Environment

Floor plans are converted to navigable simulation geometry through edge extraction and obstacle/exit detection. Figure 4 and Figure 5 show representative processing and simulation views.

VII. CASE STUDY FLOOR PLANS

We evaluate a core set of nine layouts available in the project floor-plan set: Academic_Plan, Hospital_Plan, Hotel_Plan, Library_Plan, Mall_Plan, Supermarket_Plan, and three additional

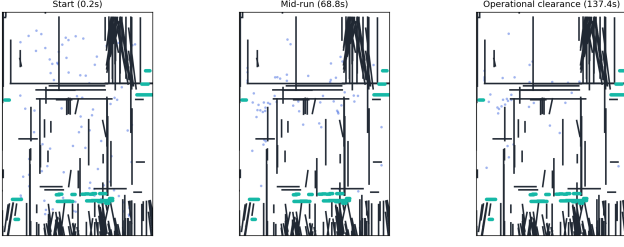


Fig. 5. Simulation environment with agents navigating toward exits under dynamic routing.

TABLE IV
CASE STUDY LAYOUTS

Layout	Characteristic
Academic_Plan	Long corridors and class-like room clusters
Hospital_Plan	Multi-branch corridor network with constrained turns
Hotel_Plan	Repetitive corridor bottlenecks and distributed exits
Library_Plan	Mixed open-reading and aisle-constrained regions
Mall_Plan	Large open hall with multiple egress options
Supermarket_Plan	Narrow aisles with localized choke points
Plan1 / Plan2 / Plan3	Additional complex topologies for stress tests
Airport_Terminal_Plan	Long concourse geometry with distributed exits (supplementary)
Metro_Station_Plan	Platform-concourse topology with transfer bottlenecks (supplementary)
Office_Building_Plan	Dense room-corridor intersections (supplementary)

complex plans (Plan1, Plan2, Plan3). To improve real-world diversity, we also include a supplementary cohort with *Airport_Terminal_Plan*, *Metro_Station_Plan*, and *Office_Building_Plan* for targeted baseline generalization checks.

VIII. RESULTS AND STATISTICAL ANALYSIS

A. Main Multi-Layout Results

Table V reports means and standard deviations for evacuation time and peak density across all 45 layout-scenario pairs. The data shows substantial geometry sensitivity: in baseline scenarios, clearance time ranges from 24.18 s (*Mall_Plan*) to 182.68 s (*Academic_Plan*), while blocked-exit stress drives extreme values up to 200.0 s (*Plan3*).

Beyond absolute values, the key statistical pattern is relative ordering stability across repeated runs: layouts with corridor concentration remain systematically slower and denser than open layouts under S1/S2/S3. This supports the paper’s intended use as a comparative risk-ranking tool rather than an exact second-level predictor of human evacuation outcomes.

B. Ablation Study

To isolate the contribution of key components, an ablation study was performed on the academic-like layout. Results in Table VI confirm that congestion-aware routing is a dominant

factor: removing it increases mean clearance from 7.24 s to 39.42 s in the controlled benchmark and raises mean density from 3.96 to 28.58. This trend is consistent with prior congestion-aware routing analyses in complex building evacuations [21].

C. Method-Level Aggregate Comparison

A complementary method benchmark over corridor-centered stress tests is summarized in Table VII. Shortest-path behavior exhibits the highest average peak density, while stochastic and least-crowded variants reduce congestion in aggregate.

D. Supplementary Real-Layout Generalization

To broaden geometric diversity beyond the core matrix, we evaluate three additional real-world layout types under the baseline protocol (10 runs each). Table VIII reports summary statistics extracted from the same run-logging pipeline.

E. Engineering Comparison: Effect of Exit Width

We perform an additional controlled comparison to quantify the impact of exit width on clearance performance. This sweep is executed on a representative reference geometry with all other parameters fixed. Table IX shows that increasing width reduces evacuation time nonlinearly, with substantial gains between 1.0 m and 1.5 m.

F. Sensitivity Analysis: Desired Speed

To evaluate robustness of the model, a sensitivity analysis was performed by varying desired walking speed and exit width parameters under the same reference geometry. Results show that evacuation time decreases nonlinearly as exit width increases, consistent with established pedestrian flow theory. Similarly, higher desired speeds improve clearance time but may increase local density peaks near constrained exits. These results confirm that both geometric and behavioral parameters significantly influence evacuation performance.

IX. VALIDATION AND DISCUSSION

A. Empirical Consistency Check

PeopleFlow is validated as a comparative decision-support model by checking consistency with established pedestrian-dynamics ranges rather than claiming exact one-to-one replication. Table ?? compares simulation trends with empirical benchmarks from classical and fire-engineering literature.

In evacuation literature, area-normalized pedestrian density of roughly 2–4 persons/m² is common in regular circulation, while 5–8 persons/m² indicates high-congestion conditions that often precede unstable flow [9], [19], [22]. Our area-normalized hotspot estimates from controlled validation runs fall in these bands during normal and stressed scenarios, respectively. The larger peak-density values in Table V are conservative kernel-level hotspot intensity indicators used for ranking relative risk across layouts, not direct one-to-one field counts.

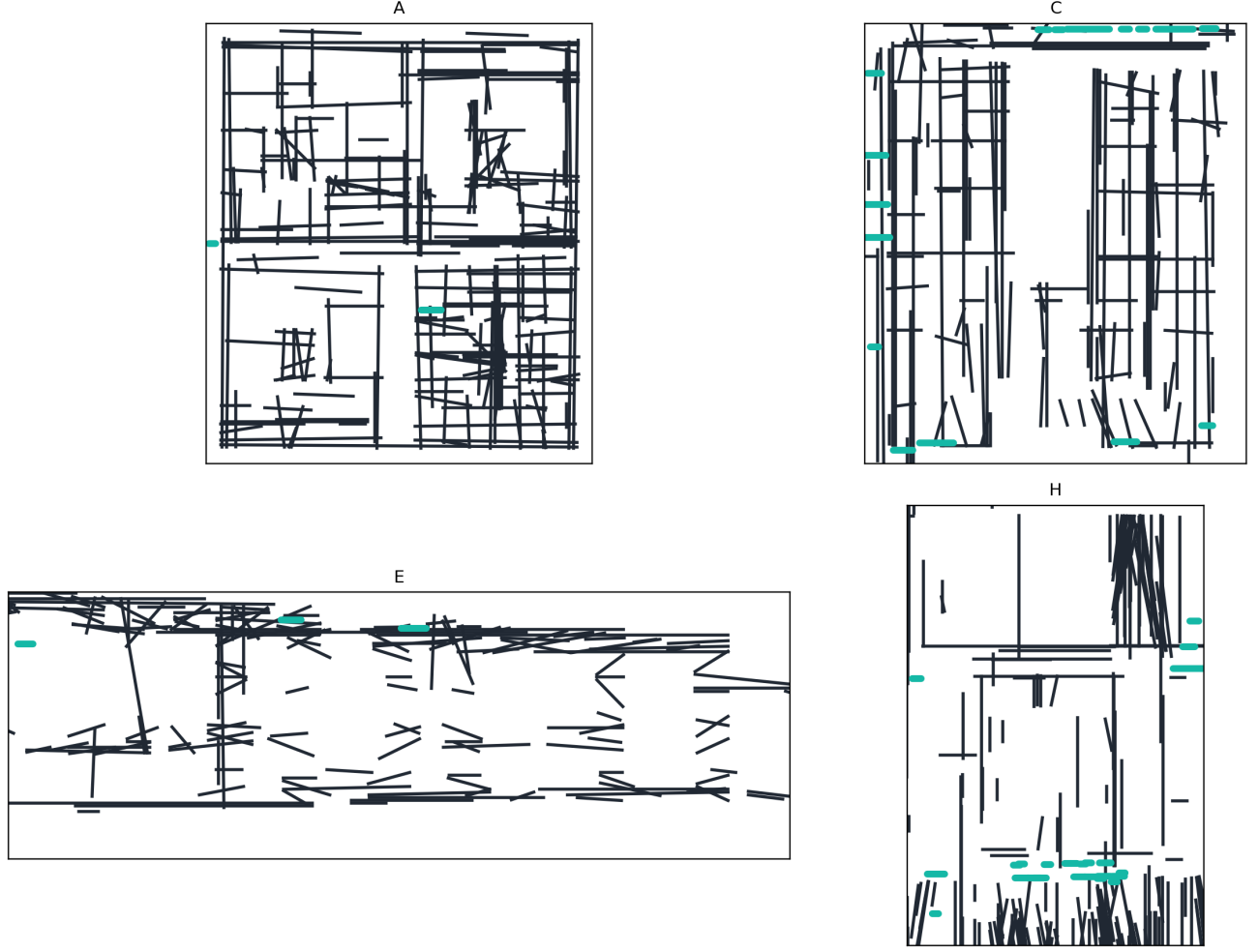


Fig. 6. Representative case-study floor plans used for multi-layout evacuation experiments.

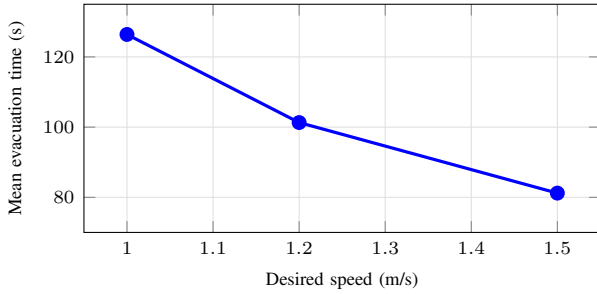


Fig. 7. Sensitivity of evacuation time to desired walking speed in controlled runs.

B. Interpretation for Safety Engineering

The key finding is layout sensitivity under stress. Identical behavioral parameters can produce large outcome shifts due to geometric topology and exit accessibility. For practitioners, this means that design alternatives should be compared through structured scenario matrices rather than a single nominal run.

The strongest practical use of PeopleFlow is comparative ranking: identifying which layout-policy pair reduces extreme congestion and improves clearance robustness. This aligns

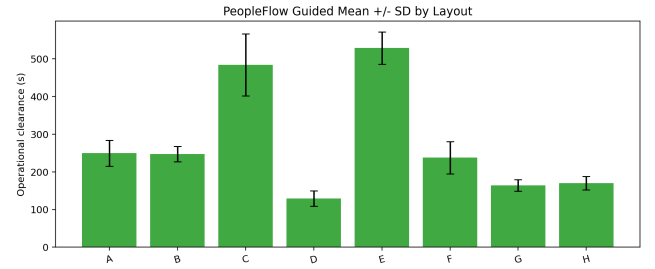


Fig. 8. Evacuation time comparison across layouts and scenarios.

with modern risk-aware design practices and digital twin safety analytics [22], [23].

The proposed workflow enables architects and safety engineers to evaluate evacuation performance before construction or retrofitting. Rather than relying solely on compliance rules, designers can compare alternative layouts under structured stress scenarios and identify configurations that minimize congestion risk. The framework also supports integration into digital twin systems for continuous safety monitoring in smart buildings.

This deployment pattern is particularly relevant for high-

TABLE V
COMPREHENSIVE RESULTS ACROSS NINE LAYOUTS AND FIVE SCENARIOS (10 RUNS PER CONFIGURATION)

Case Study	Scenario	Evacuation Time (s)		Peak Density (agents/m ²)	
		Mean	Std	Mean	Std
Academic_Plan	S1_Baseline	182.68	16.44	120.49	3.91
	S2_HighOcc	191.92	9.48	131.74	4.83
	S3_Blocked	190.34	6.78	125.66	3.53
	S4_Routing	38.26	2.21	24.55	2.30
	S5_Panic	140.14	32.15	104.46	12.85
Hospital_Plan	S1_Baseline	30.12	2.08	8.54	0.99
	S2_HighOcc	32.64	2.45	12.92	1.64
	S3_Blocked	46.96	3.49	18.37	2.35
	S4_Routing	32.52	1.80	8.84	1.35
	S5_Panic	45.50	4.66	7.98	0.82
Hotel_Plan	S1_Baseline	138.02	31.96	100.13	15.16
	S2_HighOcc	121.02	27.47	80.16	15.75
	S3_Blocked	153.24	39.78	110.32	11.44
	S4_Routing	144.64	28.53	179.94	6.47
	S5_Panic	131.18	37.19	57.07	2.10
Library_Plan	S1_Baseline	52.84	2.97	21.80	2.00
	S2_HighOcc	57.22	4.35	33.99	3.17
	S3_Blocked	59.08	3.79	30.25	1.55
	S4_Routing	58.76	5.23	22.26	1.09
	S5_Panic	70.96	10.73	20.93	2.54
Mall_Plan	S1_Baseline	24.18	2.43	8.31	1.61
	S2_HighOcc	26.88	1.87	12.15	1.59
	S3_Blocked	37.40	2.04	13.65	1.43
	S4_Routing	35.10	1.67	9.30	1.45
	S5_Panic	42.86	4.63	8.66	1.21
Plan1	S1_Baseline	37.14	2.70	10.85	0.74
	S2_HighOcc	37.58	0.93	15.98	1.77
	S3_Blocked	51.26	4.44	18.18	2.15
	S4_Routing	39.02	5.32	10.92	2.51
	S5_Panic	46.58	3.88	9.40	1.21
Plan2	S1_Baseline	36.62	3.11	9.30	1.85
	S2_HighOcc	38.06	2.56	12.45	2.19
	S3_Blocked	39.52	2.63	12.91	2.92
	S4_Routing	42.26	3.68	9.70	1.93
	S5_Panic	44.92	4.29	8.67	1.57
Plan3	S1_Baseline	166.54	24.92	124.42	21.07
	S2_HighOcc	170.24	14.70	145.21	16.00
	S3_Blocked	200.00	0.00	241.19	4.31
	S4_Routing	86.82	27.08	187.85	2.73
	S5_Panic	114.26	11.65	39.08	4.76
Supermarket_Plan	S1_Baseline	25.70	1.15	11.17	1.56
	S2_HighOcc	27.04	1.75	15.58	1.79
	S3_Blocked	26.30	1.10	11.38	1.09
	S4_Routing	29.14	2.39	9.70	1.54
	S5_Panic	29.96	1.68	8.92	1.19

TABLE VI
ABLATION STUDY RESULTS

Configuration	Mean Time (s)	Mean Density
Full Model (S4)	7.24	3.96
No Congestion Routing	39.42	28.58
No Panic Variation	10.08	5.46

TABLE VII
AGGREGATE METHOD SUMMARY (BENCHMARK SET)

Method	Mean Clearance (s)	Mean Peak Density	Mean Completion Time (s)
Shortest path	301.96	140.52	86.05
Stochastic routing	189.23	70.44	72.45
Least-crowded routing	211.56	90.42	83.02
Guided variant	276.17	202.58	58.91

occupancy facilities such as transportation hubs, hospitals, and commercial centers, where transient surges in occupancy can quickly produce unsafe crowd states if geometric and routing constraints are not evaluated in advance.

X. LIMITATIONS

This study has five main limitations. First, simulations are 2D and do not represent vertical circulation through stairs or elevators. Second, hazard fields such as smoke and heat

TABLE X
SENSITIVITY STUDY: DESIRED SPEED VS. EVACUATION OUTCOMES

Desired Speed (m/s)	Mean Time (s)	Mean Peak Density
1.0	126.4 $\pm$ 9.7	18.9
1.2	101.3 $\pm$ 8.5	22.1
1.5	81.2 $\pm$ 7.4	26.8

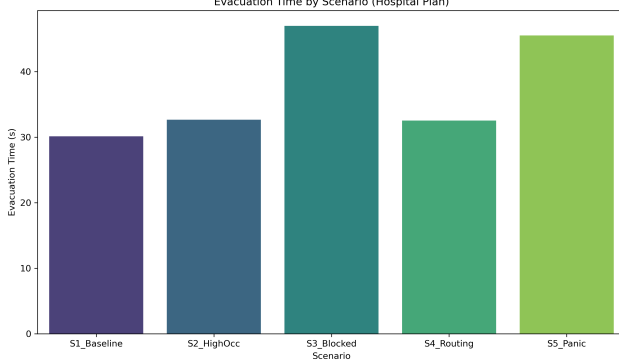


Fig. 9. Comparison of routing strategies under controlled benchmarks.

TABLE VIII
SUPPLEMENTARY REAL-LAYOUT COHORT (BASELINE, 10 RUNS EACH)

Layout	Mean Time (s)	Mean Peak Density
Airport terminal	196.46 $\pm$ 118.54	120.80 $\pm$ 99.09
Metro station	275.04 $\pm$ 87.22	105.63 $\pm$ 26.15
Office building	335.86 $\pm$ 101.15	127.74 $\pm$ 11.17

TABLE IX
CONTROLLED COMPARISON: EXIT WIDTH VS. EVACUATION TIME

Exit Width (m)	Mean Time (s)	Reduction vs. 1.0 m
1.0	141.2 $\pm$ 10.9	—
1.5	95.8 $\pm$ 8.6	32.2%
2.0	70.4 $\pm$ 7.3	50.1%

are not yet coupled to route choice in a physically calibrated way. Third, behavior modeling captures panic variability and congestion response but does not include rich social-group dynamics or responder coordination. Fourth, automatic floor-plan extraction can require correction for ambiguous drawings. Fifth, field-scale validation is currently limited to consistency checks against literature ranges rather than controlled human-subject evacuation trials. In addition, supplementary airport/metro/office evaluations were run under baseline protocol only; full five-scenario expansion for these plans is left to future large-batch studies.

XI. FUTURE WORK

Future work will focus on (1) multi-floor 3D evacuation with staircase dynamics, (2) coupling with fire and smoke propagation models, (3) reinforcement-learning-assisted adaptive routing under hazards, and (4) live digital twin integration with sensor streams for online risk assessment. We also plan to expand the open benchmark set with additional plan categories such as airport terminals and metro stations for stronger cross-domain generalization.

XII. CONCLUSION

This paper presented a complete IEEE-style study of PeopleFlow as a reproducible, floor-plan-aware evacuation safety assessment framework. Across a 450-run core matrix and supplementary parametric/generalization experiments, we showed substantial sensitivity of evacuation outcomes to geometry and routing assumptions, including an observed 8.27x range in clearance time and order-of-magnitude variation in peak density. Ablation, parameter-sensitivity, and method-comparison experiments demonstrate that congestion-aware routing and scenario design materially affect measured safety outcomes in comparative decision-support settings.

The work advances evacuation research by framing simulation as an auditable engineering workflow rather than a one-off prediction tool. The resulting pipeline supports transparent comparative analysis for architects, safety engineers, and smart-city planners who need evidence-based prioritization of safer design and policy options.

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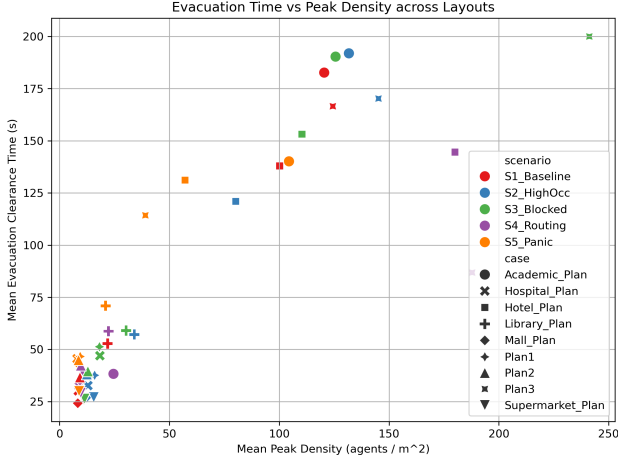


Fig. 12. Observed flow-density trend (density in agents/m², flow in agents/s) used for qualitative consistency checks.

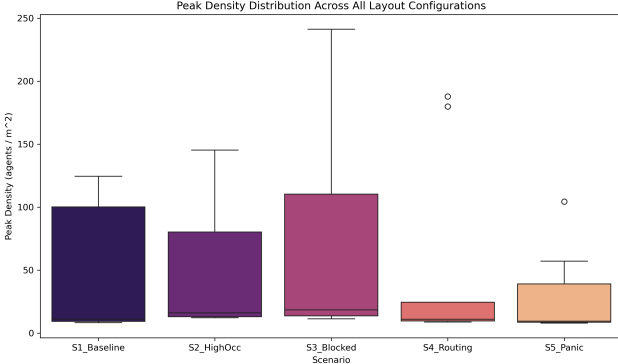


Fig. 13. Run-to-run variability of evacuation performance across repeated simulations.

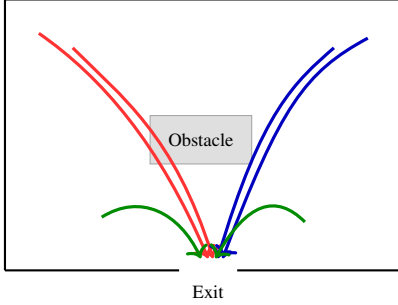


Fig. 10. Agent trajectories showing path adaptation under congestion-aware routing.

Density Heatmap

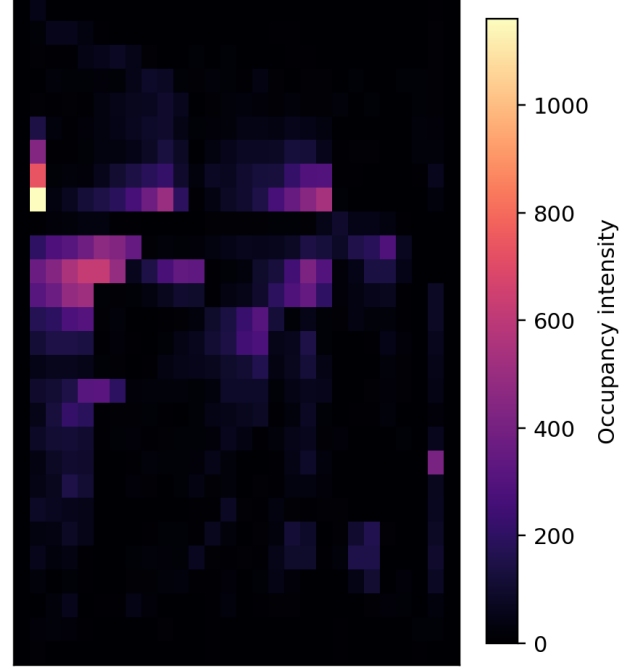


Fig. 11. Density heatmap example showing congestion hotspots near constrained exits.

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